

# Project Documentation

## C64 Keyboard Connector PCB

Project number: 224

Revision: 0

Date: 04.06.2026

# C64 Keyboard Connector PCB Rev. 0

## Module Description

### Introduction

The purpose of this PCB is to connect the C64 Keyboard (Project. No. 223 or 225) to the C64/Ultimate64 Keyboard Connector.



Figure 1: C64 Keyboard Connector PCB with ribbon cable

The connector for the ribbon cable is a MicroMatch series connector (red). They are low profile, so the cable does not collide with the side of the case when connected to a C64 long board. The C64 Keyboard has a 2x10p pin header, so the ribbon cable has two different types of connectors attached.

### Dimensions

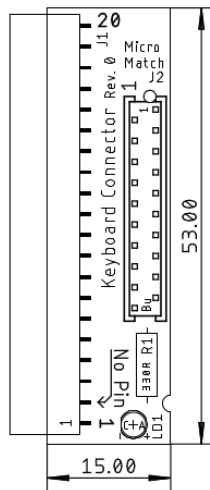


Figure 2: Dimensions of the PCB

## Pinouts

### J1 – Connector to mainboard

2.54mm pitch socket, 90°

| Pin | Signal   |
|-----|----------|
| 1   | GND      |
| 2   | No pin   |
| 3   | /RESTORE |
| 4   | +5V      |
| 5   | ROW3     |
| 6   | ROW6     |
| 7   | ROW5     |
| 8   | ROW4     |
| 9   | ROW7     |
| 10  | ROW2     |
| 11  | ROW1     |
| 12  | ROW0     |
| 13  | COL0     |
| 14  | COL6     |
| 15  | COL5     |
| 16  | COL4     |
| 17  | COL3     |
| 18  | COL2     |
| 19  | COL1     |
| 20  | COL7     |

### J2 – Ribbon Cable Connector

MicroMatch Socket, 20 pins

| Pin | Signal   |
|-----|----------|
| 1   | GND      |
| 2   | No pin   |
| 3   | /RESTORE |
| 4   | +5V      |
| 5   | ROW3     |
| 6   | ROW6     |
| 7   | ROW5     |
| 8   | ROW4     |
| 9   | ROW7     |
| 10  | ROW2     |
| 11  | ROW1     |
| 12  | ROW0     |
| 13  | COL0     |
| 14  | COL6     |
| 15  | COL5     |
| 16  | COL4     |
| 17  | COL3     |
| 18  | COL2     |
| 19  | COL1     |
| 20  | COL7     |

## Cable

The cable is a 20p ribbon cable of 45cm with one side attached to a 2x10p IDC socket with strain relief on the keyboard side and a 20 pin MicroMatch connector (male) on the side of this PCB. The cable is shown in drawing number 224-3-01-00.

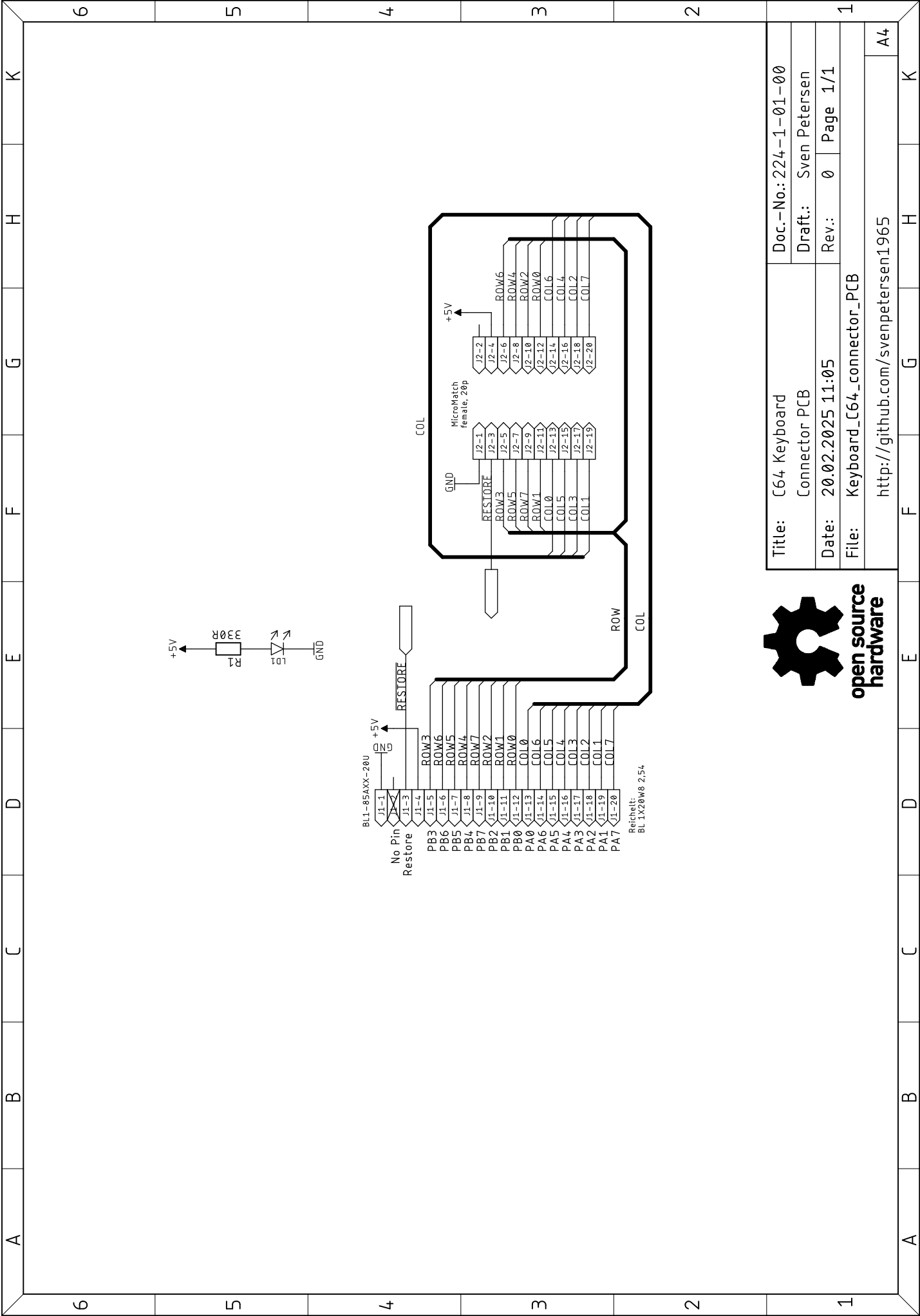
Both connector types are easy to attach, since they are both IDC (insulation-displacement contact), so you only require a vice to compress them, after the ribbon cable is positioned.

The length of 45cm is the longest possible, calculated for a C64 longboard (the keyboard connector is top left on the mainboard). For short boards, 30cm is long enough

## Revision History

### Rev. 0

- Working prototype



Title: C64 Keyboard Connector PCB

Doc.-No.: 224-1-01-00

Draft.: Sven Petersen

Date: 20.02.2025 11:05

Rev.: 0

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File: Keyboard\_C64\_connector\_PCB

<http://github.com/svenpetersen1965>

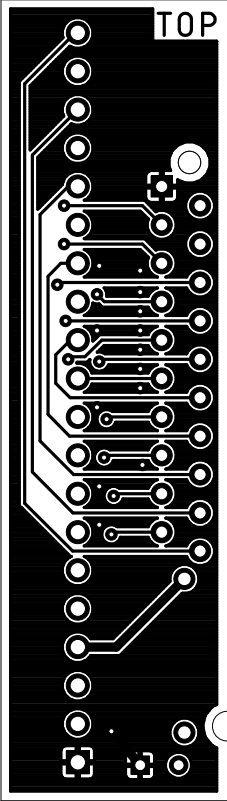
A4



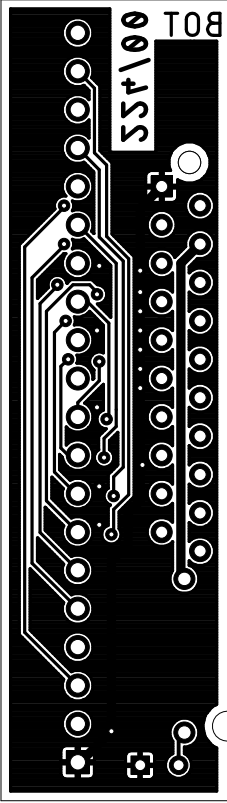
|                            |                       |              |
|----------------------------|-----------------------|--------------|
| Sven Petersen<br>2025      | Doc.-No.: 224-2-01-00 |              |
|                            | Cu: 35µm              | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB |                       |              |
| 20.02.2025 19:23           |                       | Rev.: 0      |
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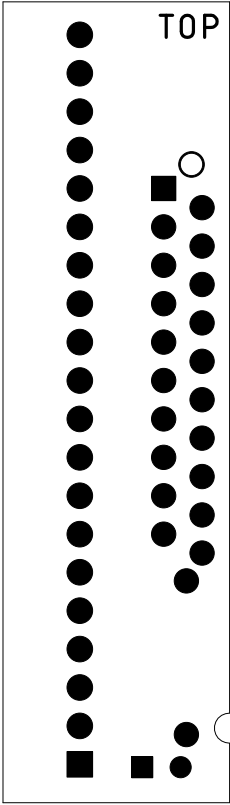
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|----------------------------|-----------------------|--------------|
| Sven Petersen<br>2025      | Doc.-No.: 224-2-01-00 |              |
|                            | Cu: 35µm              | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB |                       |              |
| 20.02.2025 19:23           |                       | Rev.: 0      |
| top                        |                       |              |



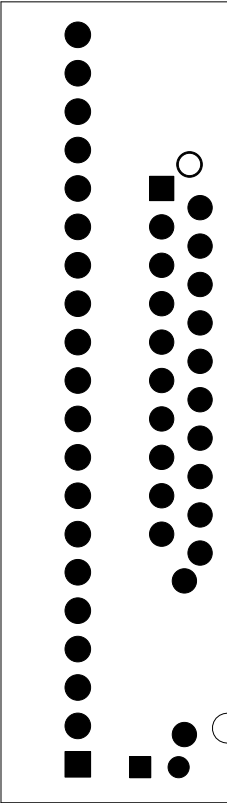
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|----------------------------|-----------------------|--------------|
| Sven Petersen<br>2025      | Doc.-No.: 224-2-01-00 |              |
|                            | Cu: 35µm              | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB |                       |              |
| 20.02.2025 19:23           |                       | Rev.: 0      |
| bottom                     |                       |              |



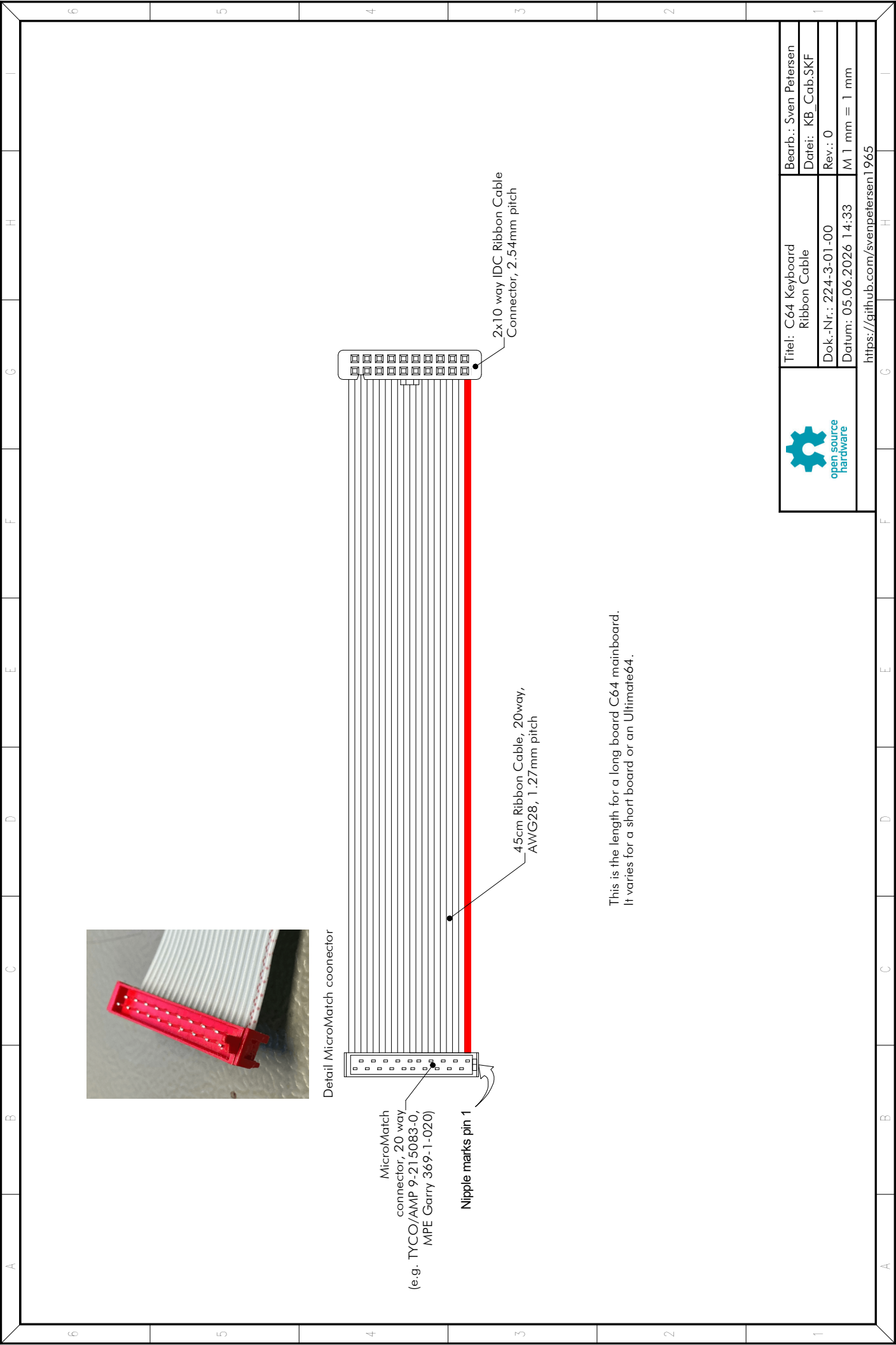
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| Sven Petersen<br>2025      | Doc.-No.: 224-2-01-00 |              |
|                            | Cu: 35µm              | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB |                       |              |
| 20.02.2025 19:23           |                       | Rev.: 0      |
| stopmask component side    |                       |              |



|                            |                       |              |
|----------------------------|-----------------------|--------------|
| Sven Petersen<br>2025      | Doc.-No.: 224-2-01-00 |              |
|                            | Cu: 35µm              | Cu-Layers: 2 |
| Keyboard_C64_connector_PCB |                       |              |
| 20.02.2025 19:23           |                       | Rev.: 0      |
| stopmask solder side       |                       |              |







|  |                                     |  |                       |
|--|-------------------------------------|--|-----------------------|
|  | Titel: C64 Keyboard Ribbon Cable    |  | Bearb.: Sven Petersen |
|  | Dok.-Nr.: 224-3-01-00               |  | Datei: KB_Cab.SKF     |
|  | Datum: 05.06.2026 14:33             |  | Rev.: 0               |
|  | https://github.com/svenpetersen1965 |  | M 1 mm = 1 mm         |

# C64 Keyboard Connector PCB Rev. 0

## Bill of Material Rev. 0.0

| Pos. | Qty | Value                                        | Footprint     | Ref.-No.   | Comment                                                                                                   |
|------|-----|----------------------------------------------|---------------|------------|-----------------------------------------------------------------------------------------------------------|
| 1    | 1   | 224-2-01-00                                  | 2 Layer       | PCB Rev. 0 | 2 layer, Cu 35μ, HASL, 53.0mm x 15.0mm, 1.6mm FR4                                                         |
|      | 1   | LED, green, 3mm                              | 3MM           | LD1        | Standard LED, 3mm, green                                                                                  |
|      | 1   | 330R                                         | R-10          | R1         | Metal film resistor, 1/4W & 10% or better                                                                 |
|      | 1   | BL1-85AXX-20U                                | BL1-85AXX-20U | J1         | 1x20 pin socket connector, 90°, e.g. MPE Garry, Reichelt BL 1x20W8 2,54                                   |
|      | 1   | 369-1-020 or 2-215079-0                      | MICMA20B      | J2         | 20 pin MicroMatch, femal, PCB-mount, e.g. MPE Garry, Reichelt MPE 369-1-020 or TE Connectivity 2-215079-0 |
|      | 1   | 45cm ribbon cable                            |               | cable      | 20 way, AWG28, 1.27mm                                                                                     |
|      | 1   | IDC header Micro Match, 20-pin               |               | cable      | Reichelt MPE 369-1-020 or TE Connectivity 9-215083-0                                                      |
|      | 1   | 20-pin IDC socket with strain relief, 2.54mm |               | cable      | e.g. Reichelt PFL 20                                                                                      |